

1

Multiple choice 1 point Question

Given a correct and efficient implementation of a list class with instance variables referencing both *head* (or *header*) and *tail* (or *trailer*), **removing** the ***last*** item (*furthest away from head*)

- ☐ is not possible in a singly-linked list, only in a doubly-linked list
- ☐ is $O(n)$ in both a singly-linked list and a doubly linked list
- ☐ is $O(1)$ in both a singly-linked list and a doubly linked list
- ☒ is $O(n)$ in a singly-linked list but $O(1)$ in a doubly-linked list

2

Multiple choice 1 point Question

Given a correct and efficient implementation of a list class with instance variables referencing both *head* (or *header*) and *tail* (or *trailer*), **removing** the ***first*** item (*furthest away from tail*)

- ☐ is $O(n)$ in both a singly-linked list and a doubly linked list
- ☐ is $O(n)$ in a singly-linked list but $O(1)$ in a doubly-linked list
- ☒ is $O(1)$ in both a singly-linked list and a doubly linked list
- ☐ is not possible in a singly-linked list, only in a doubly-linked list

3

Multiple choice 1 point Question

Given a correct and efficient implementation of a list class with instance variables referencing both *head* (or *header*) and *tail* (or *trailer*), **removing** an item at a specified **integer (index)** location in the list

- ☒ is $O(n)$ in both a singly-linked list and a doubly linked list
- ☐ is $O(n)$ in a singly-linked list but $O(1)$ in a doubly-linked list
- ☐ is $O(1)$ in both a singly-linked list and a doubly linked list
- ☐ is not possible in a singly-linked list, only in a doubly-linked list

4

Multiple choice 1 point Question

Given only a single variable referencing a particular node in a list, **adding** a new node into the list **directly in front of** the referenced node (*closer to head (or header)*)

- ☐ is $O(1)$ in both a singly-linked list and a doubly linked list
- ☒ is not possible in a singly-linked list, only in a doubly-linked list
- ☐ is $O(n)$ in both a singly-linked list and a doubly linked list
- ☐ is $O(n)$ in a singly-linked list but $O(1)$ in a doubly-linked list

5

Multiple choice 1 point Question

Given only a single variable referencing a particular node in a list, **adding** a new node into the list **directly behind** the referenced node (*closer to tail (or trailer)*)

- ☒ is $O(1)$ in both a singly-linked list and a doubly linked list
- ☐ is $O(n)$ in both a singly-linked list and a doubly linked list
- ☐ is not possible in a singly-linked list, only in a doubly-linked list
- ☐ is $O(n)$ in a singly-linked list but $O(1)$ in a doubly-linked list

